



Examiners' Report Principal Examiner Feedback

Summer 2023

Pearson Edexcel International GCSE
In Information and Communication Technology
(4IT1) Paper 01R

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IGCSE ICT 4IT1-01R

Introduction

The examination for this unit requires students to explore how current and emerging digital technologies impact on the lives of individuals, organisations and society. This is an untiered paper that has been specifically designed so that easier questions are more prevalent earlier in the paper, becoming progressively more difficult later on. This paper has been designed to be accessible, so that students of all ability ranges will find questions that are both challenging and interesting throughout. Students will find that 'command words' are used consistently in the paper to indicate the type of response expected. It is hoped that students will demonstrate the knowledge, understanding and skills required to be confident users of ICT.

Successful students often provided more detailed responses, providing more than just simple statements, including examples and reasons where expansions or explanations were required.

In the extended questions, students must demonstrate a better understanding of the requirement to 'discuss'. Students should develop their use of subject-specific and technical language, and avoid giving generic responses, responding in the context of the question.

Q01ai

Most students attempted this question with the majority being awarded the mark. The most common response tended to be 'two or more devices / computers connected together'. There were some that explained the function or purpose of a network, sharing resources, communicating with each other, etc., rather than providing a definition.

Q01bi

Many students described what a WAN was, making reference to geographical area, rather than explaining why it would be used rather than a LAN. Where students achieved a mark, it tended to be for the first mark point; few were able to then make the connection to data being stored on servers not directly connected to the LAN.

Q01bii

There was confusion about the role of a router. Often terms such as routers are used to describe devices in the home that are a combination of devices. Frequently students incorrectly described this type of device instead. Where students achieved marks for their response, they had mentioned about data (packets) / network traffic being sent / forwarded via the best or shortest route. Students need to be specific in their use of terms, e.g. packet forwarding.

Q01d

It was clear that students understood that there was some kind of list that was being checked. They were less clear about what specifically was being checked.

Q01e

Most common correct responses related to 3G / 4G, Infra-red and NFC. However, frequently students did respond with 3G and 4G as two separate points, these were in effect a repeat of the same mark point and could only be credited once. Wi-Fi and Bluetooth were sometimes given, which were repeats of the examples given in the question stem. Other incorrect responses related to 'mobile broadband', which was not creditworthy.

Q01fii

Students frequently failed to make reference to bandwidth being the maximum volume of data that can be transmitted in one second and just stated that it was the amount of data transmitted. It is important that students are more precise in their responses.

Q01gi

Various combinations of the three items in the mark scheme were seen. Where students failed to achieve one or both of the marks, they had given examples of malware rather than anti-malware. It is important that students do not give proprietary names or the names of software companies as examples, frequently this software may have more than one purpose or the company may produce more than one type of software.

Q01gii

Many students secured some marks for at least identifying the method, some then struggled to go on to describe the methods that they had identified. There were some responses that gave the examples given in the question stem (encryption, passwords and anti-malware), it is important that students read the whole question to ensure they are not repeating the given examples.

Q01h

Some students referenced drawbacks to Asher, rather than to Asher's employer and some mentioned benefits rather than drawbacks. Many students referenced the worker being distracted at home, illustrating this with examples, but then struggled to state how this would impact the employer negatively, i.e. the lack of productivity. Other popular responses related to the difficulty in monitoring employees and the data security issues.

Q02a

The majority of students were able to state that an embedded system performs a specific / dedicated / single task. There were few that referenced it forming part of another system / device. Some were able to give an example such as a dishwasher.

Unfortunately there were a few students that only gave an example, which did not actually state what is meant by an embedded system.

Q02b

Where students were successful in achieving both marks, they often referenced the monitoring of inputs and provision of outputs. Where students struggled, it was often relating to the second mark point either being too generic in the use of terms such as 'things' or 'tasks'.

Q02d

Students tended to reference the clarity of the footage or the scope of the cameras. There were some students that referenced anti-surveillance clothing. There were some responses that incorrectly referenced 'invasion of privacy'. The question related to the reliability of CCTV, not just general drawbacks or concerns relating to CCTV.

Q02eii

There were some students that gave SSD rather than HDD.

Q02ev

Most students gave at least one correct answer – 'no moving parts'. However, some students again referred to the examples given in the question stem (smaller and lighter), which could not be given any credit.

Q02gii

On the whole, this question had been attempted by most students. The majority of correct responses came from the first mark point – 'more programs being open at the same time'. There were very few examples of the second mark point. Where students did not obtain the mark, they often were not specific enough and missed referencing 'at the same time' or just stated multitasking or the computer running faster. Some students referred to the program loading more quickly, this is not the same as the program running faster once opened.

Q03a

Many students referenced 'blogs', 'wikis' and 'forums'. Some incorrectly stated social media rather than social network. Again, it is important that students avoid using proprietary names for such questions, especially as many of them offer a number of these services.

Q03c

Some students used their personal experiences of using a VPN as examples. What they had mentioned is accurate, but it didn't answer the question. Most correct responses were from the protecting privacy mark point or the award part linked to the third mark point – accessing content from other countries.

Q03d

The majority of students successfully gained at least one mark, referencing GPS. The mark relating to location services / settings was often not achieved, as their response was too vague, referring to location and not the services / settings. Some students were not specific enough and just mentioned 'tag' and not 'geotag'. The question was asking about the features of a smartphone, a few students mentioned applications / social networking allowing them to share the location and therefore were not awarded the marks as this is not a feature of the smartphone.

Q03e

Students were often limited the first part of mark point 1, that local software had more features. Some students referred to the fact no network connection was required. In both cases they then struggled to expand on these points. Most incorrect responses related to storage and privacy.

Q03fi

Most students correctly identified interference, although not specifically electromagnetic. Students struggled then to expand on this point. Frequently responses related to sharing the power with the microwave or sharing the network / bandwidth with the microwave.

Q03fii

Many students were able to reference compression, but again struggled to explain how this then related to improving the upload speed, i.e. less data being transferred. Students incorrectly stated it reduces the size of the photo, rather than the size of the file. There were some answers relating to defragmentation or reducing the storage or using internet speed boosters or PC tuning apps. Some students incorrectly mentioned using faster WiFi, a different device or updating the software.

Q03g

Students struggled to identify a drawback unique to email or different to the example given in the question stem (length of time). Some incorrect responses included reference to human error typing in the addresses or not being able to comment on the photos or privacy implications. Many answers were not unique to email, for example sharing without knowledge. Where students achieved some credit for their response, it was often related to the large file size or it being marked as spam. These points, not being linked to each other, limited them to 1 mark.

Q03hi

Although proprietary names were accepted in this response, those that did use them tended to identify two examples from the same mark point, such as two examples of cards or payment wallets. Students in future series would benefit from referring to the general terms, such as payment wallet, rather than proprietary names to avoid not achieving marks.

Q03hiii

Although students identified this was an example of phishing, they then frequently went on to describe how it can be identified as such, rather than explain the risk that is presented.

Q04ai

Again, another question that saw the responses repeat the examples given in the question stem. There were also those that gave output devices rather than input devices and did not necessarily consider the context of participating in an online meeting.

Q04b

Students struggled with this question and although gave a correct point, could not follow it up with an explanation. Students would mention the pausing of the video, but didn't reference that this meant Ella could learn at her own pace. They mentioned about asking questions, but did not expand about the comments section or that the communities could answer the questions. Again, in the question stem, the examples of anytime and anywhere were given and some responses were just a reword or repeat of those examples, and therefore not creditworthy.

Q04ci

Students were often able to identify an improvement to the search, but did not explain the fact that this made the results more relevant, specific or valid. Frequently, they suggested the results were more accurate, which is not the same as relevant, specific or valid. Phrase matching was not really seen as one of the improvements.

Q04cii

Personalisation of results, cookies and past history were frequently referred to. Profile data was not seen much and some students seemed confused with why Ella was signing in to her account. There were many answers relating to a different brand of search engine giving different results, which doesn't necessarily relate to Ella signing in.

Q04d

Most students gave good responses relating to VLEs, online workspaces or online/hosted storage.

Q04e

Where correct responses were seen, students gave a mix of positive and negative effects on the environment. Some students did not consider the context of the question and simply discussed the environmental impact of the use of devices, rather than the increased use of online services. Some students just identified negatives relating to the use of devices, rather than the effect on the environment.

Q04f

Students are becoming more confident in answering these structured response style questions; there are fewer that are left blank in each series. Some students were able to define the digital divide and then identify benefits and drawbacks in relation to the context of education. They then did struggle to go further and explain the wider benefit. At lower levels, it was clear that students confused the digital divide with digital devices in general and were therefore incorrectly talking about the advantages and disadvantages of the technology rather than the inequality in access.

Q05a

A number of students identified some kind of controller, however, the question was asking for output devices. The majority of correct responses referred to 'headphones', 'speakers' or 'screens'. Some output devices mentioned, did not necessarily relate to a games console. Students must consider the context of the question.

Q05b

This question ranged from good answers – often identifying open source and proprietary or incorrect responses relating to free or copyrighted. It is important that students have clear understanding of the types of licence.

Q05c

This question appeared to be challenging for nearly all students – they either failed to give a response, weren't able to identify the parts or confused the MAC address with an IP address. Some students identified the parts correctly, but got the labels the wrong way around and therefore did not achieve the marks. Where students were able to identify the first part related to the manufacturer, the failed to state

the second part related to the NIC, the most common response was referring to the device. Some devices can have more than one NIC.

Q05d

There is an improvement in this type of question. However, there is still a misconception that encryption stops data from being intercepted, accessed or read. Encryption doesn't prevent these from happening, it just ensures it can't be understood, should it be intercepted, accessed or read.

Q05ei

The majority of responses related to the first or second set of marks. Most responses gained 2 to 3 marks. Most students understood the benefits of using ICT to collaborate. However, collaborate is not the same as communicate, and many students simply answered, "can communicate better". Many answers relating to "working from home", rather than flexible / mobile working and students failed to mention "access the Internet anytime/anywhere". Some students mentioned high cost of equipment / resources which has nothing to do with the team's working practices. There were few answers relating to job losses / time needed to learn new skills.

Q05eii

Where students were awarded marks, they tended to only gain one mark on this question, as they identified the risk but didn't always give an explanation, or vice versa. Where students struggled, they did not consider the risk to Asher, they focussed on a risk to his data.

Q05f

Students are becoming more confident in answering these structured response style questions; there are fewer that are left blank in each series. It was clear from students that students were knowledgeable about video sharing sites, perhaps due to their experience. Centres must continue to support students with exam technique and students should look at advantages and disadvantages and then draw a conclusion. However, this particular question command word did not require this approach. Asher has already chosen to use the video sharing site and students were being asked to **justify** this choice. At a lower level, students were able to identify the benefits of using a video sharing site, some then went on to push into

Level 2 and justified these benefits, developing them more and linking to the context / scenario. Those students that scored in Level 3 were able to clearly argue a case for the justification.

Paper Summary

Based on their performance on this paper, students are offered the following advice:

- Expand and explain answers using examples and reasons, especially where more than a simple statement or list is requested
- Where a context or scenario is provided, respond with the context of the question in mind
- Read and interpret the question prompt to focus on the specific requirements and provide more relevant responses that address the critical aspects of the question.
- Develop understanding of the key pieces of network hardware and their specific role with regards to a network
- When required to give one other example/reason, ensure the response is discrete from the given example/reason
- Allocate time to plan responses to the extended questions

